Mid-semester Exam (August Session 2024)

Modern Physics – EP1108

Date: 16th October 2024 **Duration: 1.5 hrs**

Total Marks: 40

General Instructions:

1. **All the questions are compulsory.**
2. **The question paper consists of 25 questions and it is divided into three sections A, B and C.**
3. **Section A comprises of 12 questions carrying 1 mark each.**
4. **Section B comprises of 11 questions carrying 2 marks each.**
5. **Section C comprises of 2 questions carrying 3 marks each.**

Section A

1. In an experiment involving the photoelectric effect, which of the following will cause more electrons to be ejected from the metal surface?
   a) Increasing the energy of the light.
   b) Increasing the frequency of the light.
   c) Increasing the wavelength of the light.
   d) Increasing the intensity of the light.
   e) All of the above will cause more electrons to be emitted.

ANS:

(b) (d) 1

2. To what region of the electromagnetic spectrum does *975 nm* correspond?
   a) infrared region
   b) yellow part of the visible spectrum
   c) blue part of the visible spectrum
   d) ultraviolet region
   e) X-ray region

ANS:

(d) X

3. Identify which of the following statement(s) is (are) false as per Einstein's explanation of the photoelectric effect.
   a) Increasing the brightness (intensity) of incoming light at a fixed frequency increases the kinetic energy of the ejected electrons.
   b) Increasing the wavelength of incoming light increases the kinetic energy of the ejected electrons.
   c) Increasing the brightness (intensity) of incoming light increases the number of ejected electrons.
   d) Increasing the frequency of incoming light can increase the number of ejected electrons.

NS

(a), (d), (b) ✗

4. Select all the false statement(s) about electromagnetic radiation.
   a) Microwaves have shorter wavelengths than γ-rays.
   b) Red light has lower energy than blue light.
   c) Electromagnetic radiation with wavelength of *1 m* is classified as radio waves.
   d) UV rays have greater frequency than X-rays.

NS:

(a), (b) ✗

5. Select all the false statement(s) about X-ray interaction with solid
   a) X-ray photons interact with electron clouds of an atom
   b) The scattering amplitude is independent of $2q$
   c) Bragg peaks at higher angles will generally exhibit a higher intensity compared to those at lower angles
   d) X-rays are scattered in all directions but peak only seen where they interfere constructively

NS

(d) ✗

6. A beam of particles is incident from the negative x direction on a potential energy step at $x = 0$. When $x < 0$, the potential energy of the particles is zero, and for $x > 0$ the potential energy has the constant positive value U_0. In the region $x < 0$, the particles have a kinetic energy k that is smaller than U_0. What is the form of the wave function in the region $x > 0$?
   a) $Ae^{kx} + Be^{-kx}$
   b) $Ae^{ikx} + Be^{-ikx}$
   c) Ae^{kx}
   d) Ae^{-kx}

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es.

7. An atom absorbs a photon, so that the electron's total energy increases b
equal to the photon energy. In the Bohr model, the electron moves to an or
radius. what happens to the orbital speed of the electron?
(a) Increases
(b) Decreases
(c) Stays the same

ANS: (b)

8. Among the radiations emitted by atomic hydrogen we find one of energy 13.6 eV, the series limit of the Lyman series ($n = \infty$ to $n = 1$). A photon of the same energy also exists as the limit of a certain series in singly ionized helium ($Z = 2$). What is the final value of n for this photon emission?
(a) 1
(b) 2
(c) 3
(d) 4
(e) more than 4

ANS: (b)

9. In which of the following does the quantum mechanical picture of the hydrogen atom give the same result as the Bohr model?
(a) The energy levels of the electron.
(b) The orbital radius of the electron.
(c) The angular momentum of the electron.
(d) The probability distribution of the electron.

ANS: (a)

10. In the set of quantum numbers $n, l, m_l = 2, x, 1$, what are the possible values of x?
(a) $x = 0$ only
(b) $x = 0$ or 1
(c) $x = 1$ or 2
(d) $x = 1$ only
(e) $x = \pm 1$

ANS: (d)

11. Excluding cases in which the angular momentum is zero, is the length of the angul momentum vector that describes an electron in an atom always
    a) equal to z component of the angular momentum.
    b) either greater than or equal to z component of the angular momentum.
    c) always greater than the maximum possible z component of the angular momentun
    d) Always less than z component of the angular momentum.

ANS:

(b) ✗

12. A free eelctron in the motion along the x-axis has a localised wavefunction. Th uncertainty in its momentum is decreased if:
    a) The wavefunction is made narrower.
    b) The wavefunction is made broader.
    c) The wavefunction remains the same but the energy of the electron is increased
    d) The wavefunction remains the same but the energy of the electron is decreased

ANS:

(b) ✓ (1)

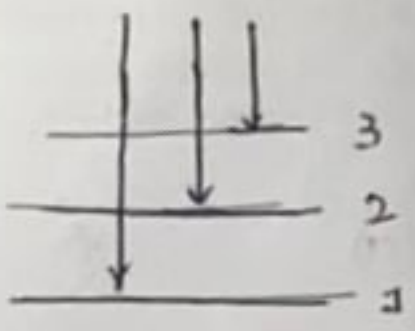
Section B

13. List out all the possible transitions which a 4p state can make.

ANS:

4P → 4S ✗
→ 3S, 3p, 3d
→ 2S, 2P
→ 1S

✗ (1)



14. The threshold frequency of potassium is *620 nm*. What is the stopping potential when light of *310 nm* is incident on potassium?

ANS:

2 eV ✓ (1)

$\frac{1240}{620} = 2$

15. At what wavelength does a room temperature ($T=16^{o}C$) ob
thermal radiation?

ANS:

$\lambda_m \cdot T = b$

$\lambda_m = b/T$

$\lambda_m = 0.0100366$

$\lambda_m \simeq 10^{-5}\ m$

16. An electromagnetic wave with the following fields is travelling along +
$\boldsymbol{E} = i E_o \sin(kz-wt)$, $\boldsymbol{B} = j B_o \sin(kz-wt)$.
Write the expression for $\boldsymbol{E}$ and $\boldsymbol{B}$ if the wave is traveling along -z axis.

ANS:

$E = -\hat{i} E_0 \sin(kz + wt)$

$B = +\hat{j} B_0 \sin(kz + wt)$

17. Two slits are *0.1mm* apart. If the incident radiation has wavelength of *530nm* an
screen is *2m* away, then what would be the distance between central maxima and
first dark fringe?

ANS:

$$\frac{\lambda D}{2d} = \frac{1}{2} \times \frac{530 \times 10^{-9} \times 2}{10^{-4}} = \frac{530 \times 10^{-5}\ m}{2}$$

18. What is the average energy of an oscillator
according to Planck's calculatio

ANS:

19. A particle in the first excited state of a one-dimensional infinite potential energy well (with $U = 0$ inside the well) has an energy of 6.0 eV. The energy of this particle in the ground state is _____.

ANS:

24 eV

X

20. A particle of mass m is confined in a rectangular box with Lx=2Ly. The state with the energy $\frac{h^2}{8mL_y^2}$ has a degeneracy of _____.

ANS:

2

X

21. A particle od mass "m" is in a cubic box of size a. The potential inside the box ($0 \leq x < a, 0 \leq y < a, 0 \leq z < a$) is zero and infinite outside. If the particle is in an eigenstate $E = \frac{14h^2}{8ma^2}$ its wave function is:

ANS:

Wave function $= \sqrt{\frac{2}{a}} \cdot \sin\left(\frac{n\pi}{a} \cdot x\right)$

$E = \frac{14h^2}{8ma^2} = \frac{7h^2}{}$

potential energy well
ergy of this particle in the

ply. The state with the

22. _____ and _____ series of triply ionised beryllium (
ANS: Give answer in nm.]

4 nm

Calculation / method

30.4 nm

$E = 13.6 \left(\frac{1}{2^2} \right)$

$E = \frac{1240}{\lambda(nm)}$

23. A Rutherford scattering experiment is set up with alpha particles
foil, and the scattered alpha particles are observed by a detector at th
θ. How does the counting rate (number of counts per second) in the de
each of the following cases? Possible answers are: (1) Increases (2)
Remains the same (4) Needs more information

a) The kinetic energy of the incident alpha particles is increased.
b) The silver foil ($Z = 47$) is replaced with gold ($Z = 79$).
c) The alpha particles are replaced with protons.
d) The thickness of the foil is decreased.

ANS:

a) Decreases (2)

b) Increases (2)

c) decreases (1)

d) Increases

24. (a) How many sets of quantum numbers n, l, m_l are possible for $n = 4$?
(b) How many of the possible sets of quantum numbers n, l, m_l for $n = 4$ have $l =$ 2?
(c) How many of the possible sets of quantum numbers n, l, m_l for $n = 4$ have $m_l =$ 0?

ANS:

a) for $n=4$, $l = 0,1,2,3$, $m_l = -3,-2,-1,0,1,2,3$

$\Rightarrow$ ~~$1+3+5+7 = 16$~~ $1+3+5+7 = 16$ //

b) for $n=4$, $l=2$ m_l can be $-2,-1,0,1,2$ $\Rightarrow$ 5 //

c) for $n=4$ & $m_l=0$, l can be $0,1,2,3$ $\Rightarrow$ 4 //

(3)

25. Which of the following is a permitted solution of the Schrodinger equation

a) $\psi(x) = A\cos kx$ $x < 0$ (Yes / ~~No~~)
b) $\psi(x) = A\sin kx$ $x > 0$ (~~Yes~~ / No)
c) $\psi(x) = Ax^{-1}e^{-kx}$ $-L \le x \le L$ (Yes / ~~No~~)
d) $\psi(x) = Asin^{-1}kx$ (Yes / ~~No~~)
e) $\psi(x) = A\tan kx$ $x > 0$ (Yes / ~~No~~)
f) $\psi(x) = \frac{Ae^{-kx}}{x}$ or $\psi(x) = Ae^{+kx}$ $x = 0$ to $x = +\infty$ and A and k are real and positive. (~~Yes~~ / No)

(2)

ANS:

(b), (f) X

X

(0)